Bavish Kulur

Edmonton, Canada
bavishkulur@gmail.com
github.com/bavishhh • bavishhh.github.io

Research Interests: Deep Reinforcement Learning, Skill Learning, Representation Learning

Education

University of Alberta Sept 2025 – July 2027 (Expected)
MSc (Thesis), Computing Science **GPA: 4.0/4.0**
Advisor: Prof. Marlos C. Machado
Coursework: Reinforcement Learning I & II, Foundations of Reasoning in LLMs, ML II

Indian Institute of Technology Bombay 2017 – 2022
Dual Degree: BTech, Mechanical Engineering; MTech, AI & Data Science **GPA: 8.84/10**
Minor in Computer Science and Engineering
Coursework: Advanced Machine Learning, Foundations of Intelligent and Learning Agents, Learning over Graphs, Linear Systems Theory, Optimization, Data Structures and Algorithms

Publication

PREFINE: Preference-Based Implicit Reward and Cost Fine-Tuning for Safety Alignment
Richa Verma, **Bavish Kulur**, Sanjay Chawla, Balaraman Ravindran
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2026. Paper Code

Work Experience

Post-Baccalaureate Research Fellow, IIT Madras Nov 2023 – Jun 2025
Robert Bosch Centre for Data Science and AI | Advisor: Prof. Balaraman Ravindran
Research projects on offline safe reinforcement learning, continual learning, and responsible AI.

Product Engineer (SDE 1), Sprinklr Gurugram, India Jul 2022 – Feb 2023
Developed backend features for Sprinklr's social-media campaign management platform, including external API integrations, data reporting, and asset management. Migrated REST APIs to GraphQL.

Software Engineering Intern, Arturo Technologies Bangalore, India Mar – Jun 2019

Teaching Assistantships

Reinforcement Learning (CMPUT 356), University of Alberta Fall 2025, 2026
File and Database Management (CMPUT 291), University of Alberta Winter 2026
Introduction to Machine Learning, NPTEL MOOC Jan – Apr 2025
Computer Programming and Utilization (CS 101), IIT Bombay Spring 2019

Selected Honors and Awards

Student Innovation Grant, AI and Robotics Technology Park (ARTPARK), Indian Institute of Science 2021
Selected for a semester exchange at the National University of Singapore 2020
Kishore Vaigyanik Protsahan Yojana (KVPY) award, Government of India 2017
Qualified in the Regional Mathematical Olympiad and attended the Indian National Mathematics Olympiad training camp at Indian Statistical Institute Bangalore 2015

Selected Research Projects

Online Skill Discovery with Laplacian Representations Jan 2026 – Present
University of Alberta | Advisor: Prof. Marlos C. Machado

- Investigating how Laplacian representations can be used as intrinsic reward functions for learning diverse and far reaching skills in continuous control environments.
- Identified failure modes of existing methods for learning Laplacian representations when learning online and proposed modified objective.

Gradient Ascent based Q-Learning Algorithm for Continuous Action Spaces Sept – Dec 2025
University of Alberta | Course project with Prof. Martha White | Report Code

- Proposed a value-based RL algorithm for continuous actions that initializes action search with a learned proposal function and refines it through gradient ascent to construct TD targets.
- Evaluated against GreedyAC and Soft Actor-Critic (SAC), outperforming both baselines in the project experiments.

Spectral Bias of Deep Learning in Lazy Regime Sept – Dec 2025
University of Alberta | Course project with Prof. Csaba Szepesvari | Report

- Reviewed theoretical work on the implicit bias of gradient descent, focusing on minimum-degree bias, spectral bias, and neural tangent kernel (NTK) theory.
- Connected results across different learning regimes to explain how minimum-degree bias emerges as a special case of spectral bias.

PREFINE: Safe RL through Direct Preference Optimization 2024 – 2025
Advisors: Prof. Balaraman Ravindran (IIT Madras) and Dr. Sanjay Chawla (QCRI, Qatar)

- Coauthored a method adapting Direct Preference Optimization (DPO) to trajectory-level preferences, fine-tuning pretrained continuous-control policies for safety while retaining reward performance.
- The method uses policy-sampled counterfactual actions for preference comparisons and achieves over 60% fewer constraint violations and catastrophic failures compared to baselines.

Fairness Evaluation of Distilled Language Models 2024 – 2025
Advisors: Prof. Balaraman Ravindran (IIT Madras) and Prof. Karthik Sheshadri (NIT AP)

- Investigated the impact of knowledge distillation from large language models on fairness in task-specific distilled models.
- Compared gender bias across forward and reverse KL distillation objectives, finding that mode-seeking behaviour of reverse KL generally resulted in lower bias.

Technical Skills

Programming and systems: Python, C++, Docker, Slurm, Git, Linux

Machine learning: JAX, PyTorch, Flax, Optax, Optuna, Hydra, Hugging Face Transformers

RL and simulation: Gymnasium, MuJoCo, Brax

Volunteering

Convener, Web and Coding Club, IIT Bombay 2018 – 2019

Developer and Founding Member, InstiApp, Developers Community, IIT Bombay 2018 – 2019